Cyclistic Bike-Share Analysis: Google Data Analytics Capstone Project

Prepared by: Sabbir Ahmed

Email: sabbirstat11@gmail.com

LinkedIn: www.linkedin.com/in/sabbir-ahmed-54b884404

Date : 01-07-2026

Tools Used: Python, Pandas, Tableau Public

Abstract

This capstone project, completed as part of the Google Data Analytics Professional Certificate program, analyzes the riding behavior of Cyclistic bike-share users to support the company's goal of converting casual riders into annual members. Using twelve months of Divvy trip data (June 2025–May 2026), comprising more than 5.8 million cleaned records across 22 columns, the study applies the Ask, Prepare, Process, Analyze, Share, and Act framework of the Google Data Analytics process. Data cleaning, feature engineering, and exploratory analysis were performed in Python with the Pandas library, and key findings were visualized through an interactive Tableau Public dashboard. The results show clear behavioral differences between the two user groups: annual members ride more frequently overall, follow a consistent weekday pattern with distinct morning and evening commuting peaks, and start and end their trips mainly at stations in downtown business areas. Casual riders, by contrast, take considerably longer trips, ride predominantly on weekends and during the summer months, and favor stations near tourist and recreational destinations such as Navy Pier and Millennium Park. Both groups show a strong preference for electric bikes. Based on these insights, the report recommends seasonal and weekend-targeted marketing campaigns, location-based promotions at high-traffic recreational stations, and messaging that emphasizes the cost savings and everyday convenience of an annual membership. These data-driven recommendations are intended to guide Cyclistic's marketing strategy toward increasing membership conversion and long-term revenue growth.

Contents

Chapter 1: Introduction

Cyclistic is a bike-share company that provides an affordable, convenient, and environmentally friendly transportation service for residents and visitors. To meet the diverse needs of its customers, the company offers several pricing options, including single-ride passes, day passes, and annual memberships. While both customer groups contribute to Cyclistic's success, annual members are considered more valuable because they generate consistent long-term revenue and are more likely to use the service regularly.

This case study aims to analyze the differences in riding behavior between annual members and casual riders using **12 months of Divvy trip data (June 2025 – May 2026)**. The analysis examines key aspects of rider behavior, including ride frequency, average ride duration, bike type preferences, and riding patterns by weekday, month, hour, and station usage. These insights help identify how each customer group uses the service and reveal opportunities to improve membership conversion.

The project follows the Google Data Analytics process and uses **Python** for data cleaning and analysis and **Tableau Public** for interactive data visualization. The findings presented in this report are intended to support data-driven business decisions and provide practical recommendations for converting more casual riders into annual members.

Chapter 2: Ask – Business Task

Business Background

Cyclistic is a bike-share company that offers flexible transportation services through single-ride passes, day passes, and annual memberships. While both casual riders and annual members contribute to the company's success, annual members are more valuable because they provide a stable source of long-term revenue. As a result, increasing annual memberships has become one of Cyclistic's key business goals.

Business Problem

Cyclistic's marketing team wants to understand how annual members and casual riders use the bike-sharing service differently. By identifying these behavioral differences, the company can develop effective marketing strategies to encourage more casual riders to purchase annual memberships.

Business Task

Analyze **12 months of historical bike-share data (June 2025 – May 2026)** to compare the riding behavior of annual members and casual riders. The analysis examines ride frequency, ride duration, bike type preferences, riding patterns by weekday, month, hour, and station usage to identify trends that distinguish the two customer groups.

Objective

The primary objective of this case study is to generate **data-driven insights** that explain the differences between annual members and casual riders and to provide actionable recommendations that help Cyclistic convert more casual riders into annual members.

Chapter 3: Methodology

3.1: Prepare – Data Source

Dataset

This analysis uses the **Divvy Trip Data**, a publicly available dataset provided for the Google Data Analytics Capstone Project. The dataset contains trip-level information such as ride start and end times, bike types, station names, geographic coordinates, and rider categories (annual members and casual riders).

Data Collection Period

The dataset includes **12 months of historical trip data**, covering the period from **June 2025 to May 2026**.

Data Size

After combining all monthly datasets and completing the data cleaning process, the final dataset contained **5,848,674 rows and 22 columns**.

Data Credibility and Limitations

The dataset contains real-world bike-sharing trip records and does not include personally identifiable information (PII), making it suitable for analysis. However, it does not provide demographic information or the purpose of each trip, so conclusions are based solely on observed riding behavior.

3.2: Process – Data Cleaning

The data preparation process was performed using **Python** and the **Pandas** library. Twelve monthly Divvy trip datasets (**June 2025 to May 2026**) were imported and combined into a single dataset for analysis.

The dataset was then cleaned to improve data quality and ensure reliable results. Duplicate records were checked, missing and inconsistent values were handled where necessary, and records with invalid or negative ride durations were removed to maintain data accuracy.

As part of the **feature engineering** process, several new columns were created, including **ride_duration, month, day_of_week, hour, month_name, day, year,** and **hour_label**. These new features made it easier to analyze riding patterns across different time periods and compare the behavior of annual members and casual riders.

After completing the data cleaning and feature engineering process, the final dataset contained **5,848,674 rows and 22 columns**, providing a reliable dataset for exploratory data analysis and visualization.

Chapter 4: Analysis

Analysis 1: Member vs. Casual Ride Count

Business Question

How does the total number of rides differ between annual members and casual riders?

 Visualization

Insert Figure 1: Member vs. Casual Ride Count

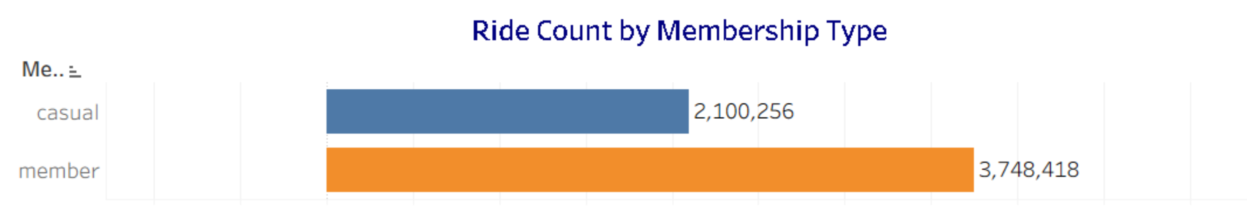

Figure 1. Total ride count comparison between annual members and casual riders.

Key Findings

User Type	Total Rides	Percentage
Member	3,748,418	64.09%
Casual	2,100,256	35.91%

- Annual members completed **3,748,418 rides**, representing **64.09%** of all rides.
- Casual riders completed **2,100,256 rides**, accounting for **35.91%** of total rides.
- Annual members generated nearly two-thirds of all rides during the study period.

Interpretation

The results show that annual members are the primary users of Cyclistic's bike-sharing service, generating the majority of total rides throughout the study period. This suggests that members rely on the service more regularly, likely for daily transportation and commuting. Although casual riders account for a smaller share of rides, they still represent more than one-third of total usage, making them an important customer segment with strong potential for business growth.

Business Recommendation

- Continue maintaining high service quality to retain existing annual members.
- Develop targeted marketing campaigns to encourage casual riders to purchase annual memberships.
- Since casual riders represent over one-third of all rides, converting even a small percentage of them into annual members could significantly increase long-term membership revenue.
- Further analysis of ride duration, weekday, monthly, and hourly riding patterns can help identify the best opportunities for membership conversion

Analysis 2: Rideable Type Preference by User Type

Business Question

How do bike type preferences differ between annual members and casual riders?

 Visualization

Insert Figure 2: Rideable Type Preference by User Type

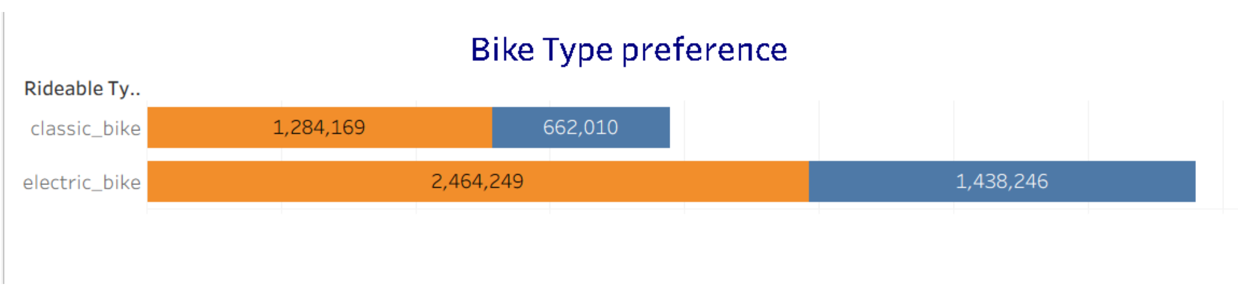

Figure 2. Comparison of classic bike and electric bike usage between annual members and casual riders.

Key Findings

User Type	Classic Bike (Count)	Classic Bike (%)	Electric Bike (Count)	Electric Bike (%)
Casual	662,010	31.52%	1,438,246	68.48%
Member	1,284,169	34.26%	2,464,249	65.74%

- Electric bikes were the most preferred bike type for both user groups.
- **68.48%** of casual riders used electric bikes, compared to **65.74%** of annual members.
- Casual riders showed a slightly stronger preference for electric bikes than annual members.
- Classic bikes were used less frequently by both groups.

Interpretation

The results show that electric bikes are the preferred choice for both annual members and casual riders. However, casual riders have a slightly higher preference for electric bikes than members. Although the difference is small, it may suggest that casual riders value the convenience and ease of using electric bikes, especially for occasional or recreational trips. This indicates that electric bikes play an important role in attracting both new and existing riders.

Business Recommendation

- Continue expanding and maintaining the electric bike fleet to meet rider demand.
- Highlight the convenience and accessibility of electric bikes in marketing campaigns targeting casual riders.
- Develop promotional campaigns that encourage casual riders who frequently use electric bikes to purchase an annual membership.
- Further analysis of ride duration, trip timing, and travel purpose can help explain why casual riders prefer electric bikes.

Analysis 3: Average Ride Duration by User Type

Business Question

How does the average ride duration differ between annual members and casual riders?

□ **Visualization**

Insert Figure 3: Average Ride Duration by User Type

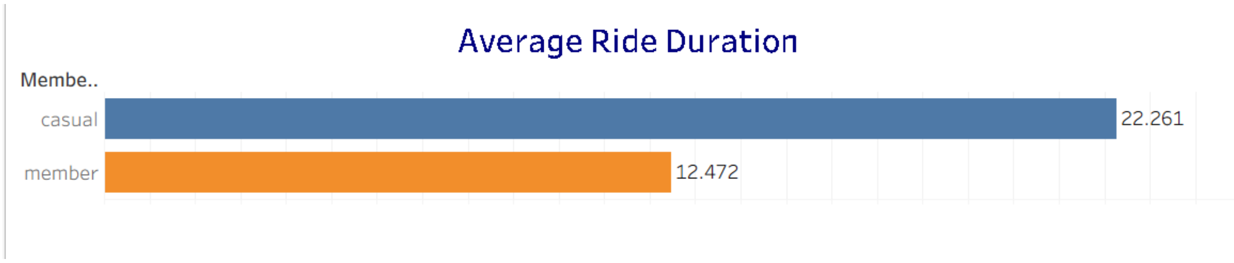

Figure 3. Comparison of average ride duration between annual members and casual riders.

Key Findings

User Type	Mean (Minutes)	Median (Minutes)	Min	Max
Casual	22.26	11.27	0.0008	1574.90
Member	12.47	8.59	0.0012	1559.90

- Casual riders had a significantly longer average ride duration than annual members.
- The average ride duration for casual riders (**22.26 minutes**) was nearly twice that of annual members (**12.47 minutes**).
- The median ride duration was also higher for casual riders (**11.27 minutes**) than for members (**8.59 minutes**).
- The large difference between the mean and median suggests that a small number of very long rides increased the average ride duration, especially for casual riders.

Interpretation

The results indicate that casual riders generally spend more time on each trip than annual members. This suggests that casual riders are more likely to use Cyclistic bikes for leisure, sightseeing, or recreational purposes. In contrast, annual members tend to take shorter and more regular trips, which is consistent with daily commuting or routine transportation. These differences highlight distinct riding behaviors between the two user groups.

Business Recommendation

- Promote the cost savings of an annual membership to casual riders who frequently take longer rides.
- Design marketing campaigns that show how an annual membership can provide better value for regular bike users.
- Create seasonal or weekend membership offers that appeal to leisure riders.
- Use these insights together with weekday, monthly, and hourly riding patterns to develop more targeted membership campaigns.

Analysis 4: Ride Pattern by Day of the Week

Business Question

Do annual members and casual riders use Cyclistic bikes differently across the days of the week?

 Visualization

Insert Figure 4: Ride Pattern by Day of the Week

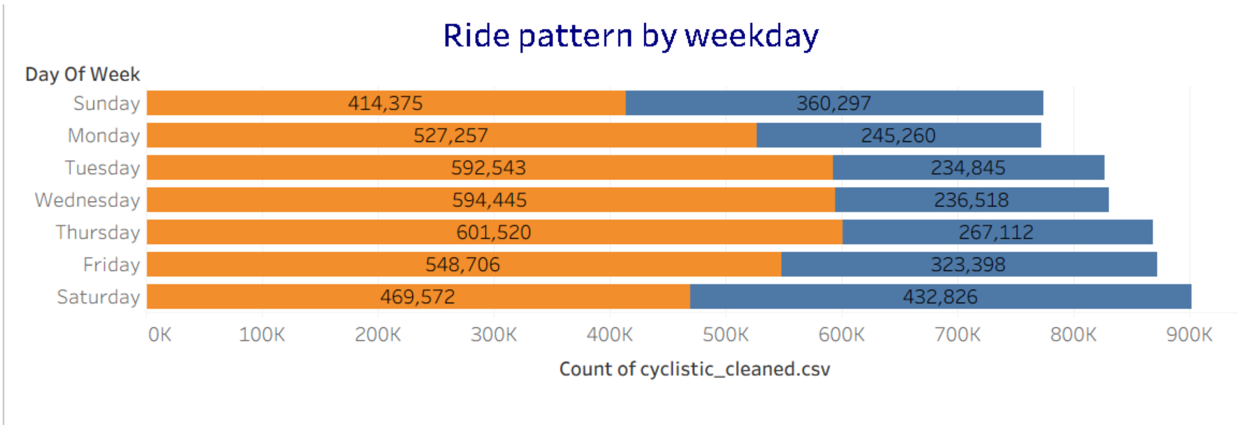

Figure 4. Weekly ride distribution of annual members and casual riders.

Key Findings

Day	Casual Rides	Casual %	Member Rides	Member %
Monday	245,260	11.68%	527,257	14.07%
Tuesday	234,845	11.18%	592,543	15.81%
Wednesday	236,518	11.26%	594,445	15.86%
Thursday	267,112	12.72%	601,520	16.05%
Friday	323,398	15.40%	548,706	14.64%
Saturday	432,826	20.61%	469,572	12.53%
Sunday	360,297	17.15%	414,375	11.05%

- Annual members recorded the highest number of rides between **Tuesday and Thursday**, with **Thursday** having the highest ride count (**601,520 rides**).

- Casual riders were most active on **weekends**, especially on **Saturday**, with **432,826 rides**.
- Weekend rides made up a much larger share of total rides for casual riders than for annual members.
- Members showed a more consistent riding pattern throughout the workweek, while casual riders clearly preferred weekends.

Interpretation

The weekly riding patterns reveal two different ways Cyclistic bikes are used. Annual members ride more frequently during weekdays, suggesting that they mainly use the service for commuting and other regular daily trips. In contrast, casual riders are most active on weekends, indicating that they are more likely to use the bikes for leisure, recreation, or sightseeing. These patterns suggest that annual members depend on Cyclistic as part of their daily routine, while casual riders use the service mainly for occasional activities.

Business Recommendation

- Launch weekend marketing campaigns that encourage casual riders to upgrade to an annual membership.
- Promote the benefits of using an annual membership for both weekday commuting and weekend leisure trips.
- Highlight the long-term cost savings and unlimited ride benefits of annual memberships to frequent weekend riders.
- Develop targeted promotions that encourage casual riders to ride more often during weekdays.

Analysis 5: Ride Pattern by Month

Business Question

Do annual members and casual riders use Cyclistic bikes differently throughout the year?

□ **Visualization**

Insert Figure 5: Ride Pattern by Month

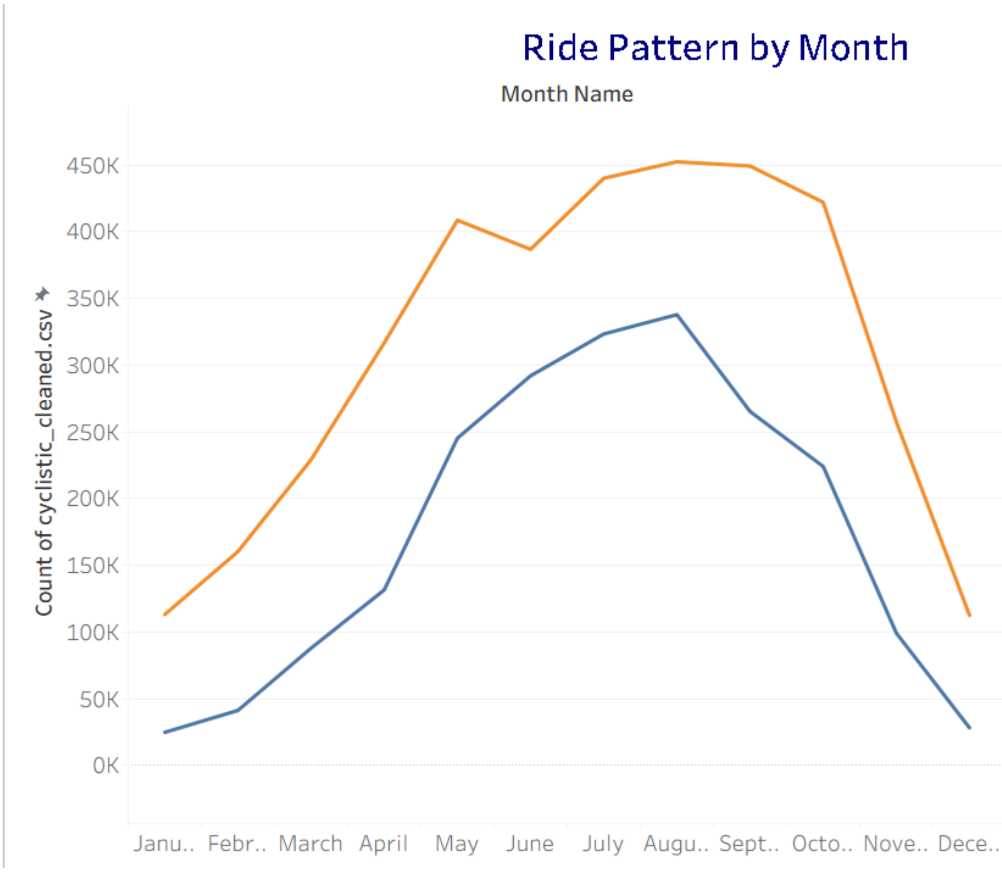

Figure 5. Monthly ride distribution of annual members and casual riders.

Key Findings

Month	Casual Rides	Casual %	Member Rides	Member %
June	292,006	13.90%	386,795	10.32%
July	323,352	15.40%	440,106	11.74%
August	337,878	16.09%	452,439	12.07%
September	265,268	12.63%	449,294	11.99%
October	224,038	10.67%	422,058	11.26%
November	99,082	4.72%	257,401	6.87%

Month	Casual Rides	Casual %	Member Rides	Member %
December	28,090	1.34%	112,455	3.00%
January	24,740	1.18%	113,042	3.02%
February	41,173	1.96%	160,281	4.28%
March	87,850	4.18%	229,165	6.11%
April	131,526	6.26%	316,761	8.45%
May	245,253	11.68%	408,621	10.90%

- Casual riders reached their highest activity in **August (16.09%)**, followed by **July (15.40%)** and **June (13.90%)**.
- Annual members also recorded more rides during the summer months, but their monthly riding pattern remained more balanced throughout the year.
- Ride activity for both user groups declined during the winter months (**December–February**).
- Casual riders showed much stronger seasonal variation than annual members.
- Annual members continued using Cyclistic bikes throughout the colder months, while casual rider activity dropped significantly.

Interpretation

The monthly riding patterns suggest that seasonal changes have a greater impact on casual riders than on annual members. Casual riders are most active during the warmer months, likely using Cyclistic bikes for leisure, outdoor activities, and tourism. Their ride frequency decreases sharply during winter. In contrast, annual members continue riding throughout the year with a more consistent usage pattern, suggesting that they rely on Cyclistic for regular transportation rather than only seasonal recreation.

Business Recommendation

- Launch membership campaigns between **May and August**, when casual rider activity is at its highest.
- Promote the long-term cost savings and convenience of an annual membership during the peak riding season.
- Offer limited-time summer membership discounts to encourage casual riders to become annual members.
- Use seasonal marketing campaigns to convert highly active summer riders into year-round members.

Analysis 6: Ride Pattern by Hour of the Day

Business Question

Do annual members and casual riders use Cyclistic bikes at different times of the day?

 Visualization

Insert Figure 6: Ride Pattern by Hour of the Day

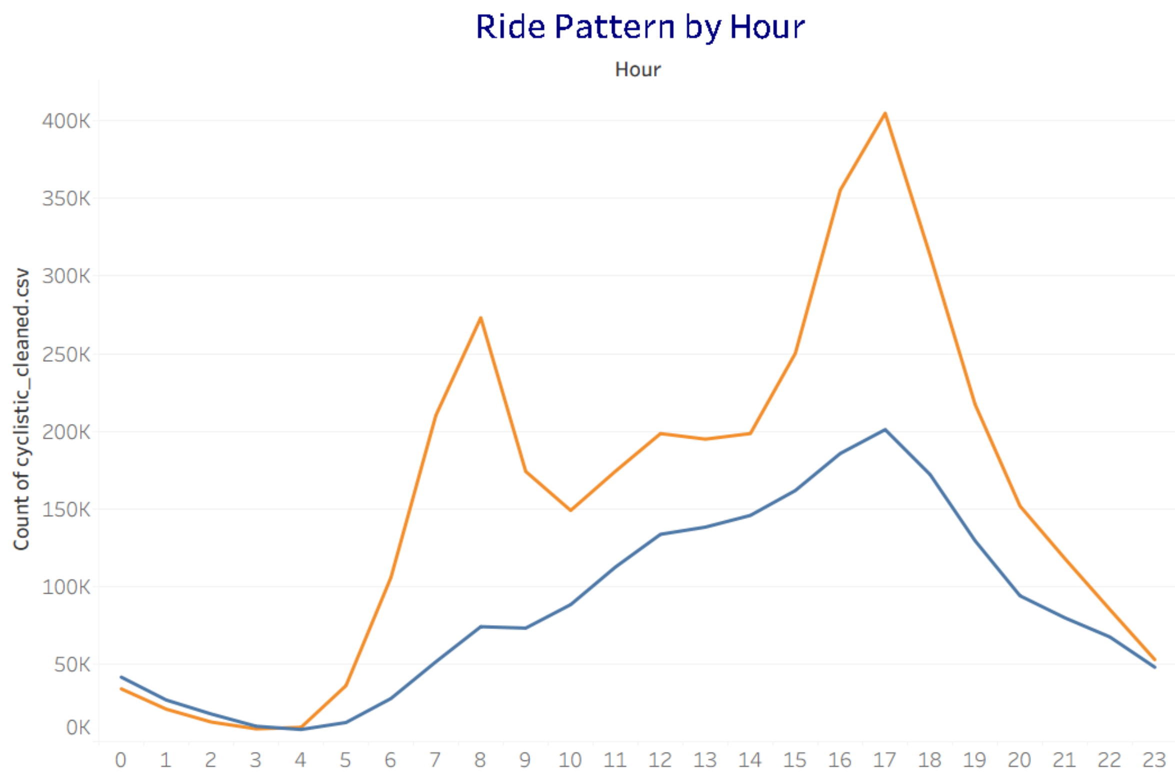

Figure 6. Hourly ride distribution of annual members and casual riders.

Key Findings

Time of Day	Casual Rides	Casual %	Member Rides	Member %
12 AM – 01 AM	41,533	1.98%	33,975	0.91%

Time of Day	Casual Rides	Casual %	Member Rides	Member %
01 AM – 02 AM	26,805	1.28%	20,912	0.56%
02 AM – 03 AM	17,803	0.85%	12,624	0.34%
03 AM – 04 AM	9,896	0.47%	8,229	0.22%
04 AM – 05 AM	7,812	0.37%	9,196	0.25%
05 AM – 06 AM	12,308	0.59%	36,092	0.96%
06 AM – 07 AM	27,767	1.32%	105,767	2.82%
07 AM – 08 AM	51,443	2.45%	210,231	5.61%
08 AM – 09 AM	74,061	3.53%	273,154	7.29%
09 AM – 10 AM	73,137	3.48%	174,094	4.64%
10 AM – 11 AM	88,268	4.20%	148,957	3.97%
11 AM – 12 PM	112,574	5.36%	174,324	4.65%
12 PM – 01 PM	133,594	6.36%	198,504	5.30%
01 PM – 02 PM	138,233	6.58%	194,916	5.20%
02 PM – 03 PM	145,775	6.94%	198,527	5.30%
03 PM – 04 PM	161,794	7.70%	250,203	6.67%
04 PM – 05 PM	185,686	8.84%	355,388	9.48%
05 PM – 06 PM	201,063	9.57%	404,939	10.80%
06 PM – 07 PM	172,095	8.19%	313,326	8.36%
07 PM – 08 PM	129,476	6.16%	217,347	5.80%
08 PM – 09 PM	93,979	4.47%	151,822	4.05%
09 PM – 10 PM	79,766	3.80%	117,914	3.15%

Time of Day	Casual Rides	Casual %	Member Rides	Member %
10 PM – 11 PM	67,451	3.21%	85,154	2.27%
11 PM – 12 AM	47,937	2.28%	52,823	1.41%

- Annual members showed a clear commuting pattern, with ride activity increasing rapidly between **7:00 AM and 9:00 AM**.
- Member rides reached their highest level between **5:00 PM and 6:00 PM**, accounting for **10.80%** of all member rides.
- Casual riders were active throughout the afternoon, with their highest ride activity also occurring between **5:00 PM and 6:00 PM (9.57%)**.
- Compared with members, casual riders had relatively higher activity during the late-night and early-morning hours (**12:00 AM–3:00 AM**).
- Casual riders also recorded longer average ride durations throughout the day, reinforcing that they generally take longer trips than annual members.

Interpretation

The hourly riding pattern highlights a clear behavioral difference between annual members and casual riders. Annual members mainly ride during the morning and evening commuting hours, suggesting that they use Cyclistic bikes as part of their daily travel routine for work or school. In contrast, casual riders have a more evenly distributed riding pattern throughout the day. Their activity increases from late morning, remains high during the afternoon, and continues into the evening. They are also relatively more active during late-night hours and tend to take longer trips, indicating that their rides are more likely to be for leisure or recreational purposes.

Business Recommendation

- Promote annual memberships as a convenient and cost-effective option for daily commuting.
- Launch marketing campaigns during peak commuting hours to encourage frequent casual riders to become annual members.
- Highlight benefits such as unlimited rides, lower long-term costs, and convenience for both work and leisure travel.
- Combine these insights with weekday and monthly riding patterns to create more targeted membership campaigns.

Analysis 7: Top Start Stations by User Type

Business Question

Do annual members and casual riders begin their rides from different locations?

□ **Visualization**

Insert Figure 7: Top Start Stations by User Type

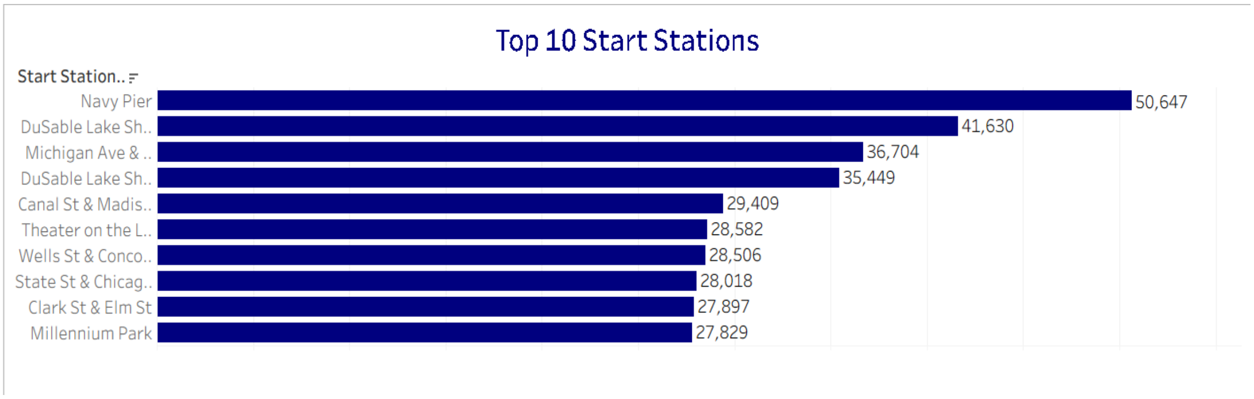

Figure 7. Top 10 start stations used by annual members and casual riders.

Key Findings

Rank	Member Start Station	Member Rides	Casual Start Station	Casual Rides
1	Canal St & Madison St	22,518	Navy Pier	38,895
2	Clinton St & Washington Blvd	20,867	DuSable Lake Shore Dr & Monroe St	32,879
3	State St & Chicago Ave	20,624	Michigan Ave & Oak St	23,243
4	Clinton St & Madison St	20,441	DuSable Lake Shore Dr & North Blvd	20,293
5	Kingsbury St & Kinzie St	20,367	Millennium Park	19,127
6	Wells St & Elm St	19,456	Shedd Aquarium	17,179
7	Wells St & Concord Ln	19,205	Theater on the Lake	16,511
8	Clark St & Elm St	18,939	Dusable Harbor	15,579
9	Clinton St & Jackson Blvd	18,818	Streeter Dr & Grand Ave	15,230
10	Wells St & Huron St	18,395	Michigan Ave & 8th St	11,237

- Annual members most frequently started their rides from stations located in downtown business and commuting areas.
- Casual riders mainly began their trips from stations near tourist attractions, parks, and waterfront destinations.
- There was **no overlap** between the top 10 start stations of annual members and casual riders.
- The most popular casual rider station, **Navy Pier**, recorded **38,895 rides**, compared with **22,518 rides** at the most popular member station, **Canal St & Madison St**.

Interpretation

The analysis shows that annual members and casual riders begin their trips from different types of locations. Annual members mainly start their rides from stations in business districts and near public transportation, suggesting that they use Cyclistic bikes for commuting and daily travel. In contrast, casual riders most often begin their trips from popular tourist attractions and recreational areas, indicating that their rides are primarily for leisure and sightseeing. These location-based patterns further support the differences in riding behavior identified in the previous analyses.

Business Recommendation

- Promote annual memberships at stations with high casual rider activity, especially near tourist attractions and recreational areas.
- Place membership advertisements and promotional offers at popular locations such as **Navy Pier, Millennium Park,** and **Shedd Aquarium**.
- Encourage casual riders to upgrade to an annual membership by highlighting the convenience and long-term cost savings of becoming a member.
- Focus location-based marketing efforts on high-traffic tourist destinations to improve membership conversion rates.

Analysis 8: Top End Stations by User Type

Business Question

Do annual members and casual riders end their rides at different locations?

□ **Visualization**

Insert Figure 8: Top End Stations by User Type

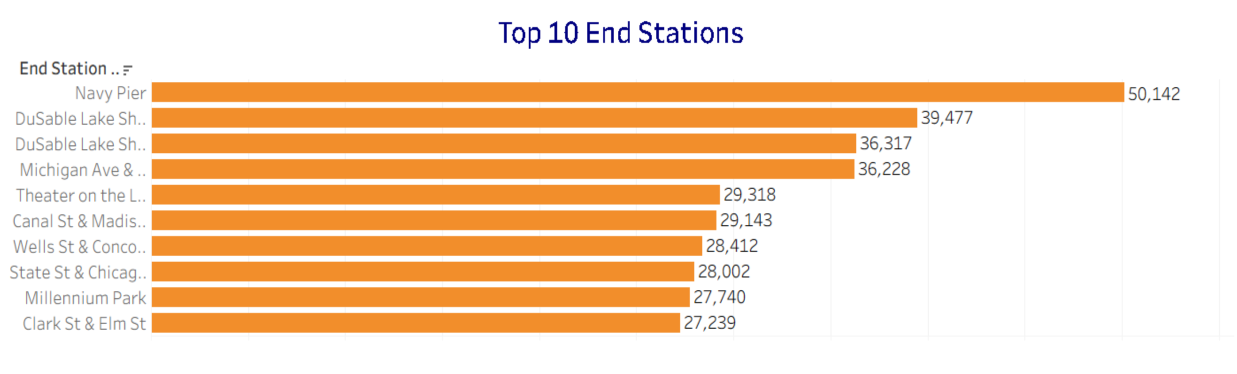

Figure 8. Top 10 end stations used by annual members and casual riders.

Key Findings

Rank	**Member End Station**	**Member Rides**	**Casual End Station**	**Casual Rides**
1	Canal St & Madison St	21,962	Navy Pier	39,821
2	State St & Chicago Ave	21,023	DuSable Lake Shore Dr & Monroe St	30,140
3	Clinton St & Madison St	20,792	Michigan Ave & Oak St	23,411
4	Kingsbury St & Kinzie St	20,482	DuSable Lake Shore Dr & North Blvd	22,470
5	Clinton St & Washington Blvd	20,194	Millennium Park	19,945
6	Clark St & Elm St	19,587	Theater on the Lake	17,818
7	Wells St & Concord Ln	19,269	Streeter Dr & Grand Ave	15,588
8	Wells St & Elm St	19,077	Shedd Aquarium	15,051
9	Clinton St & Jackson Blvd	18,333	Dusable Harbor	13,468
10	Wells St & Huron St	17,596	Montrose Harbor	10,755

- Annual members most frequently ended their rides at stations located in downtown business and commuting areas.
- Casual riders mainly ended their trips at tourist attractions, waterfront destinations, and recreational locations.
- Similar to the start station analysis, there was **no overlap** between the top 10 end stations of annual members and casual riders.
- **Navy Pier** remained the most popular destination for casual riders with **39,821 rides**, while **Canal St & Madison St** was the most common destination for annual members with **21,962 rides**.

Interpretation

The end station analysis confirms the behavioral differences identified throughout the previous analyses. Annual members typically end their rides in business districts and commercial areas, indicating that Cyclistic bikes are mainly used for commuting and routine transportation. In contrast, casual riders frequently finish their trips at tourist attractions and recreational destinations, suggesting that their rides are primarily leisure-oriented. The similarity between the start and end station patterns provides strong evidence that the two user groups use Cyclistic bikes for different purposes.

Business Recommendation

- Focus membership marketing campaigns at popular recreational and tourist destinations where casual riders frequently end their trips.
- Place QR codes, digital advertisements, and membership promotions at high-traffic locations to encourage sign-ups.
- Highlight the convenience and long-term value of an annual membership to casual riders while they are actively using the service.
- Use location-based marketing strategies to improve membership conversion among frequent casual riders.

Chapter 5: Interactive Dashboard (Share)

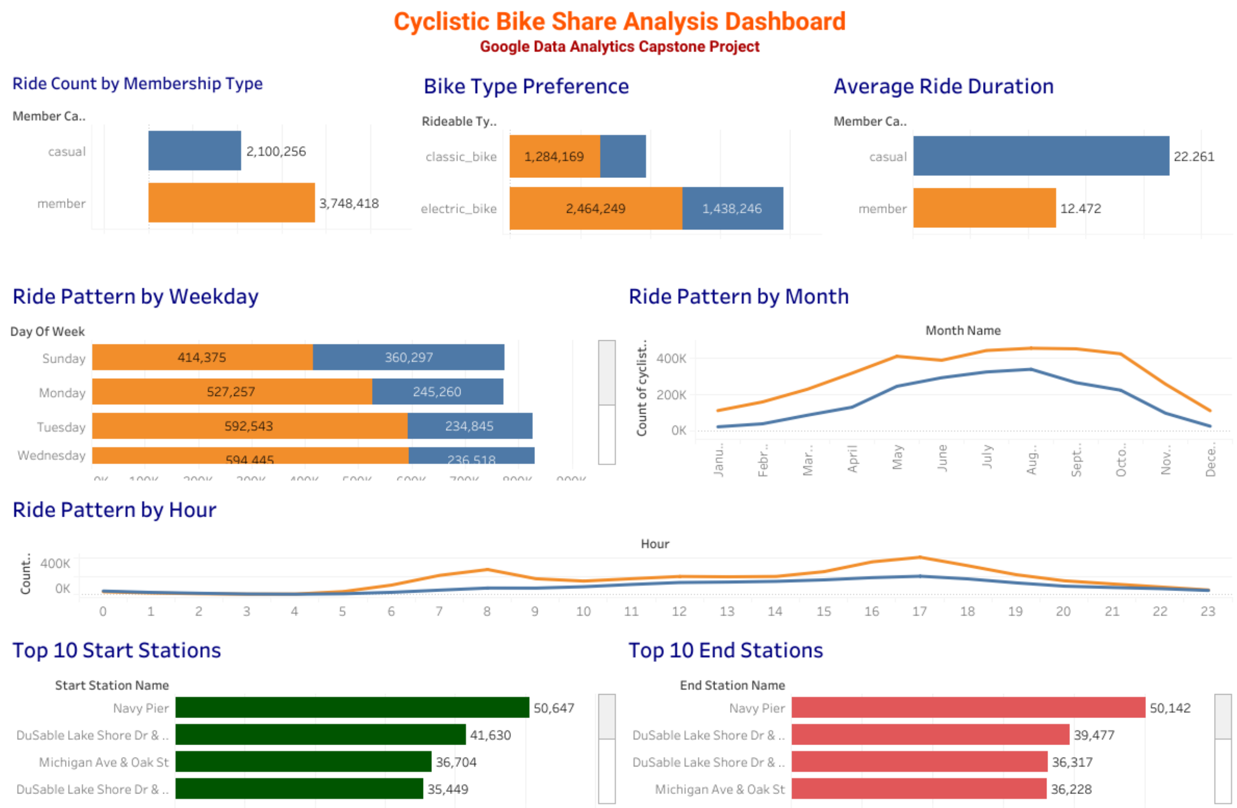

Figure 9. Interactive Tableau dashboard summarizing the key insights of the Cyclistic bike-share analysis.

An interactive dashboard was developed in **Tableau Public** to present the key findings of this project. The dashboard allows users to compare the riding behavior of annual members and casual riders through interactive visualizations, including ride count, ride duration, bike type preference, riding patterns by weekday, month, and hour, as well as the most popular start and end stations.

Tableau Public Dashboard:

https://public.tableau.com/app/profile/sabbir.ahmed3169/viz/dashboardcyclistic22/CyclisticDashboard1

Chapter 6: Final Recommendations

Based on the analysis, annual members and casual riders use Cyclistic bikes for different purposes. Annual members mainly use the service for daily commuting, while casual riders are more likely to ride for leisure, recreation, and tourism. To increase annual memberships, Cyclistic should focus on the following strategies:

1. Launch Seasonal Membership Campaigns

Casual rider activity is highest between **May and August**, especially during the summer months. Cyclistic should run membership campaigns during this period when casual riders are most active and more likely to consider becoming annual members.

2. Target Weekend Riders

Casual riders show a strong preference for riding on **weekends**, particularly on **Saturdays**. Special weekend promotions, limited-time membership discounts, and weekend events can encourage these riders to switch to an annual membership.

3. Promote the Benefits of Daily Membership Use

Many casual riders may not realize that an annual membership can also be useful for everyday transportation. Marketing campaigns should highlight benefits such as **unlimited rides, lower long-term costs, convenience, and flexibility** for both commuting and leisure trips.

4. Focus Marketing at Tourist and Recreational Locations

The start and end station analyses show that casual riders frequently use stations near **Navy Pier, Millennium Park, Shedd Aquarium,** and other popular recreational areas. Cyclistic should place membership advertisements, QR codes, and promotional materials at these high-traffic locations to reach potential members while they are actively using the service.

5. Promote Electric Bikes in Membership Campaigns

Electric bikes are the most preferred bike type for both annual members and casual riders. Cyclistic can use this preference in its marketing by highlighting the comfort, convenience, and accessibility of electric bikes as part of the benefits of an annual membership.

6. Encourage Frequent Casual Riders to Become Members

Casual riders take **longer trips** and use the service regularly during peak seasons and weekends. Personalized offers, free trial memberships, or limited-time discounts could encourage these frequent users to become annual members and increase long-term customer retention.

Chapter 7: Conclusion

This case study explored the differences between annual members and casual riders using Cyclistic bike-share data from **June 2025 to May 2026**. The analysis revealed clear differences in riding behavior between the two user groups. Annual members primarily use Cyclistic bikes for daily commuting, with more frequent rides during weekdays, peak commuting hours, and throughout the year. In contrast, casual riders tend to take longer trips and are more active on weekends, during the summer months, and around tourist and recreational locations.

These findings suggest that casual riders represent a valuable opportunity for membership growth. By targeting casual riders during peak seasons, weekends, and at popular recreational destinations, Cyclistic can increase the likelihood of converting them into annual members. Promoting the convenience, flexibility, and long-term cost savings of an annual membership through targeted marketing campaigns can help strengthen customer loyalty and support sustainable business growth.

References

1. Divvy Bikes. (2026). *Divvy trip data* [Data set]. Motivate International Inc. https://divvybikes.com/system-data

2. Google. (n.d.). *Google Data Analytics Professional Certificate*. Coursera. https://www.coursera.org/professional-certificates/google-data-analytics

3. McKinney, W. (2010). Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference* (pp. 56–61). https://pandas.pydata.org/

4. Python Software Foundation. (n.d.). *Python language reference*. https://www.python.org/

5. Tableau Software, LLC. (n.d.). *Tableau Public*. https://public.tableau.com/